Course notes

Divisibility in $\mathbb{Z}$

Definition · Divisor, multiple. Let a and b be two integers. We say that a *divides* b, written $a \mid b$, if there is an integer k such that $b = ka$. We also say that a is a *divisor* of b, or that b is a *multiple* of a.

Property · Properties of divisibility. Let a, b, c be integers.

1. If $a \mid b$ and $b \mid c$, then $a \mid c$.
2. If $a \mid b$ and $a \mid c$, then $a \mid bu + cv$ for all integers u and v.
3. If $a \mid b$ and $b \neq 0$, then $|a| \leq |b|$.

Proof. 1. $b = ka$ and $c = k'b$ give $c = (k'k)a$. 2. $b = ka$ and $c = k'a$ give $bu + cv = (ku + k'v)a$. 3. $b = ka$ with $k \neq 0$, so $|k| \geq 1$ and $|b| = |k|\,|a| \geq |a|$. □

Property · Euclidean division. Let $a \in \mathbb{Z}$ and $b \in \mathbb{N}$, $b \geq 1$. There is a unique pair of integers (q, r) such that $a = bq + r$ and $0 \leq r < b$. We call q the *quotient* and r the *remainder*.

Proof. *Existence.* Let $q = \left\lfloor \frac{a}{b} \right\rfloor$, the largest integer less than or equal to $\frac{a}{b}$. Then $q \leq \frac{a}{b} < q + 1$, so $bq \leq a < bq + b$, and $r = a - bq$ satisfies $0 \leq r < b$.

Uniqueness. If $a = bq + r = bq' + r'$ with $0 \leq r, r' < b$, then $b(q - q') = r' - r$ and $-b < r' - r < b$. The only multiple of b strictly between $-b$ and b is 0, so $r = r'$ and then $q = q'$. □

Watch out. The remainder is always nonnegative: $-47 = 5 \times (-10) + 3$, so the division of -47 by 5 has quotient -10 and remainder 3, not -9 and -2.

Congruences

Definition · Congruence modulo n. Let $n \geq 2$ be an integer. Two integers a and b are *congruent modulo* n, written $a \equiv b \pmod{n}$, if $n \mid a - b$.

Property · Congruence and remainder. $a \equiv b \pmod{n}$ if and only if a and b have the same remainder in the Euclidean division by n. In particular, every integer is congruent modulo n to its remainder, which belongs to $\{0, \ldots, n-1\}$.

Proof. Write $a = nq + r$ and $b = nq' + r'$ with $0 \leq r, r' < n$. Then $a - b = n(q - q') + (r - r')$ and $-n < r - r' < n$. So $n \mid a - b$ if and only if $n \mid r - r'$, if and only if $r = r'$. □

Property · Compatibility with the operations. If $a \equiv b \pmod{n}$ and $c \equiv d \pmod{n}$, then

$$a + c \equiv b + d, \qquad a - c \equiv b - d, \qquad ac \equiv bd, \qquad a^k \equiv b^k \pmod{n}$$

for every natural number k. The relation is also transitive: if $a \equiv b$ and $b \equiv c \pmod{n}$, then $a \equiv c \pmod{n}$.

Proof. Sum: $(a + c) - (b + d) = (a - b) + (c - d)$ is a sum of two multiples of n. Product: $ac - bd = a(c - d) + d(a - b)$, a multiple of n. Power: induction on k from the product. Transitivity: $a - c = (a - b) + (b - c)$. □

Watch out. A congruence is not simplified like an equality: $2 \times 3 \equiv 2 \times 1 \pmod 4$, but $3 \not\equiv 1 \pmod 4$. Division is possible only in specific cases (chapter 7).

Method · Remainder of a large power. Look for a power congruent to 1 (or to -1), then use the Euclidean division of the exponent. Example: $2^3 = 8 \equiv 1 \pmod 7$ and $2{,}026 = 3 \times 675 + 1$, so $2^{2026} = \left(2^3\right)^{675} \times 2 \equiv 1^{675} \times 2 = 2 \pmod 7$.

Method · Table of congruences. To study an expression modulo n, draw up a table giving its value modulo n for each possible remainder of x. Example: modulo 4, $x \equiv 0, 1, 2, 3$ gives $x^2 \equiv 0, 1, 0, 1$. A square is therefore congruent to 0 or 1 modulo 4.

Method · Divisibility test for 9. If $N = \sum_{k=0}^{p} a_k 10^k$ is written with the digits $a_0, \ldots, a_p$, since $10 \equiv 1 \pmod 9$, we have $10^k \equiv 1$ and $N \equiv a_0 + a_1 + \cdots + a_p \pmod 9$. An integer is divisible by 9 if and only if the sum of its digits is.